

Project Proposal: HR Management System

1. Project Overview

This project aims to develop a scalable Human Resource (HR) Management System using an Oracle SQL database backend and a Bootstrap-based HTML user interface. The system manages core employee data, department assignments, job roles, salary histories, and attendance tracking.

2. Database Schema Design (Oracle SQL)

The database consists of 5 highly normalized tables ensuring data integrity:

Table Name	Primary Key	Foreign Keys / Attributes
DEPARTMENTS	dept_id (NUMBER)	dept_name (VARCHAR)
JOBS	job_id (NUMBER)	job_title (VARCHAR), min_salary (NUMBER), max_salary (NUMBER)
EMPLOYEES	emp_id (NUMBER)	first_name (VARCHAR), last_name (VARCHAR), email (VARCHAR), hire_date (DATE), dept_id (FK), job_id (FK)
SALARY_HISTORY	history_id (NUMBER)	emp_id (FK), Component_Salary (NUMBER), effective_date (DATE)
ATTENDANCE_LOGS	log_id (NUMBER)	emp_id (FK), log_date (DATE), status (VARCHAR - e.g., 'Present', 'Absent')

3. User Interface (Bootstrap HTML)

- **Dashboard:** Overview cards summarizing total employees and departments using Bootstrap .card classes.
- **Forms:** Simple, clean input fields using .form-control to add new employees or log attendance.
- **Data Tables:** Display directories using responsive table views.

4. Scalability & Future Enhancements

The schema is built to easily scale without disrupting core records. Future modules can include:

- **Leave Management:** A separate LEAVE_REQUESTS table linked via emp_id.
- **Performance Appraisals:** A PERFORMANCE_REVIEWS table to track structural feedback.